MS-thesis Defense
'Study of Finite Abelian Extensions of $\mathbb{Q}$'

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Let $\zeta_n := e^{2\pi i/n} \in \mathbb{C}$ denote the primitive n^{th} root of unity. A cyclotomic extension of $\mathbb{Q}$ is a finite abelian extension of $\mathbb{Q}$ which is of the form $\mathbb{Q}(\zeta_n)$, for some positive integer n.

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The Kronecker-Weber theorem

Any finite abelian extension K of $\mathbb{Q}$ is contained in some cyclotomic extension of $\mathbb{Q}$.

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Some Examples

- The three quadratic extensions $\mathbb{Q}(i), \mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$ are all contained in the cyclotomic extension $\mathbb{Q}(\zeta_8)$.
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  This is because, if $\mathbb{Q}(\sqrt[3]{5}) \subset \mathbb{Q}(\zeta_n)$ for some positive integer n, then, as the Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ is abelian (isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$), by the fundamental theorem of Galois theory, every sub-extension of $\mathbb{Q}(\zeta_n)$ is normal over $\mathbb{Q}$.

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  The splitting field of $X^3 - 5$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{5}, \zeta_3)$ and note that $\mathrm{Gal}(\mathbb{Q}(\sqrt[3]{5}, \zeta_3)/\mathbb{Q})$ is non-abelian.

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  But, we see that $\mathbb{Q}(\sqrt[3]{5})/\mathbb{Q}$ is not normal over $\mathbb{Q}$. Thus, $\sqrt[3]{5}$ is **not** a cyclotomic integer. □

Another Example

Let ζ_p denote a primitive p^{th} root of unity. Consider the quadratic Gauss sum

$$g := \sum_{t=0}^{p-1} \left(\frac{t}{p}\right) \zeta_p^t \in \mathbb{Q}(\zeta_p),$$

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As $g^2 = (-1)^{(p-1)/2} p$ and moreover,

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Thus,

- $\mathbb{Q}(\sqrt{p}) \subset \mathbb{Q}(\zeta_p)$ if $p \equiv 1 \mod 4$
- $\mathbb{Q}(i\sqrt{p}) \subset \mathbb{Q}(\zeta_p)$ if $p \equiv 3 \mod 4$

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Sketch of the proof of K-W

One of the key ingredients in Hilbert's proof of the KW-theorem is the tool from the geometry of numbers - *the Minkowski bound*[2], and its corollary, the *Hermite-Minkowski theorem*.

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Minkowski's Bound

Let K be an algebraic number field of degree n over $\mathbb{Q}$. Then, each ideal class contains an ideal $\mathfrak{a}$ satisfying such that

$$N_{K/\mathbb{Q}}(\mathfrak{a}) \leq \frac{n!}{n^n} \cdot \left(\frac{4}{\pi}\right)^{r_2} \cdot \sqrt{|d_K|}$$

where r_2 is the number of pairs of distinct complex (non-real) embeddings of K, $n = [K : \mathbb{Q}]$ and d_K is the (absolute) discriminant of K.

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The Hermite-Minkowski theorem

For any number field $K \neq \mathbb{Q}$, its the absolute discriminant d_K is not equal to ± 1.

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Let K be a number field. As the rational primes that ramify in K are precisely the ones that divide the discriminant d_K, we conclude that at least one rational prime ramifies in K. i.e.,
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Now, using the theory of higher ramification groups, we show that K is contained in a succession of number fields of the form $F_n(\zeta_n)$, where $F_n \subset K$, such that in each succession, the number of rational primes that ramify can be made smaller in exchange for adjoining appropriate roots of unity.

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Note that each F_n is hence, necessarily abelian over $\mathbb{Q}$. Eventually, we end up with an unramified abelian extension F_m over $\mathbb{Q}$, for some $m \in \mathbb{N}$. Thus, $F_m = \mathbb{Q}$ as $\mathbb{Q}$ has no proper unramified extensions. Thus, $K \subset \mathbb{Q}(\zeta_m)$. $\triangle$

Class Field Theory

Class field theory (CFT) is the description of abelian extensions of global fields and local fields. In our project, we have taken up the study of global class field theory for $\mathbb{Q}$ and have mostly followed the book by Janusz[3].

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- CFT also presents a rule (via generalized congruences) for classifying *splitting primes* in abelian extensions of number fields, presents a *general reciprocity law* and also answers questions about primes of the form $x^2 + ny^2$, for integers x and y, and some infinitely many integers n.

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*For any abelian extension L/K of number fields, and a modulus $\mathfrak{m}$ for K divisible by the ramified primes over L, the Artin map $\varphi^{\mathfrak{m}}_{L/K} : \mathscr{I}^{\mathfrak{m}}_K \to \mathrm{Gal}(L/K)$ is **surjective** [4],*

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induced by the Artin map $\varphi^{\mathfrak{m}}_{L/K} : \mathscr{I}^{\mathfrak{m}}_K \to \mathrm{Gal}(L/K)$. If that is the case, then the field L is called the **ray class field** of K for the modulus $\mathfrak{m}$ and is denoted by $K(\mathfrak{m})$.

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We finally state a generalization of the Kronecker-Weber theorem to every number field K.

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A generalization of the Kronecker-Weber theorem

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For each modulus $\mathfrak{m}$ for K, the ray class field $K(\mathfrak{m})$ of K exists. And every finite abelian extension L/K is contained in some ray class field.

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  However, for most number fields $K \neq \mathbb{Q}$, there is no systematic way to write down its ray class fields and the fields $K(\mathfrak{m})$ are generally neither cyclotomic extensions of K, nor contained in cyclotomic extensions of K.

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- Kronecker's *liebster Jugendtraum*[5] was to generalize the Kronecker-Weber theorem to imaginary quadratic extensions of $\mathbb{Q}$, which is now known via the theory of complex multiplication on elliptic curves. Using the special values of elliptic and modular functions, Kronecker was able to generate abelian extensions of imaginary quadratic fields. However, for any general number field, this problem (known as *Hilbert's* 12th *problem*) is also open as of today.

[5] "dearest dream of his youth"

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- Now, $\mathbb{Z}/(p-1)\mathbb{Z}$ necessarily contains an index 3 subgroup, which corresponds to a degree 3 subfield of $\mathbb{Q}(\zeta_p)$ under the Fundamental theorem of Galois theory.
- Thus, $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ both contain a degree 3 abelian extension of $\mathbb{Q}$.

Some non-trivial examples

Question

How do we find the cyclotomic extensions which contain, say, degree 3 (abelian) extensions of $\mathbb{Q}$?

- We consider the primes $p \equiv 1 \mod 3$ (which are infinitely many in number, say, as an application of Dirichlet's theorem on primes in arithmetic progressions).
- We now consider the cyclotomic extension $\mathbb{Q}(\zeta_p)/\mathbb{Q}$ which has a Galois group isomorphic to $(\mathbb{Z}/p\mathbb{Z})^\times \simeq \mathbb{Z}/(p-1)\mathbb{Z}$.
- Now, $\mathbb{Z}/(p-1)\mathbb{Z}$ necessarily contains an index 3 subgroup, which corresponds to a degree 3 subfield of $\mathbb{Q}(\zeta_p)$ under the Fundamental theorem of Galois theory.
- Thus, $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ both contain a degree 3 abelian extension of $\mathbb{Q}$.

Question

Now, what about $\mathbb{Q}(\zeta_7, \zeta_{13})$? How many degree 3 subextensions does $\mathbb{Q}(\zeta_7, \zeta_{13})$ have? Can we characterize them by finding their splitting polynomials over $\mathbb{Q}$, their ramified primes, and/or their ideal class groups?

SOLUTION

- $\mathrm{Gal}(\mathbb{Q}(\zeta_7, \zeta_{13})) \simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.

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- As we are required to calculate the number of index 3 subgroups in $\mathrm{Gal}(\mathbb{Q}(\zeta_7, \zeta_{13}))$, the question boils down[6] to finding the degree 3 quotients and subgroups of $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.

[6] If H is a degree subgroup of $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, then $\frac{G}{\mathbb{Z}/2\mathbb{Z} \times H \times \mathbb{Z}/4\mathbb{Z}}$ is a subgroup of index 3 in G.

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- A quick calculation (of the number of 'lines', i.e., $1-$dimensional linear subspaces of the vector space $\mathbb{F}_p^2$ over $\mathbb{F}_p$) shows that there are exactly $p+1$ subgroups of order p in the group $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.

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- We now ignore the two 'trivial' degree 3 subfields of $\mathbb{Q}(\zeta_7, \zeta_{13})$ which correspond to the ones contained in $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ respectively, and look for the other two.

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- We now ignore the two 'trivial' degree 3 subfields of $\mathbb{Q}(\zeta_7, \zeta_{13})$ which correspond to the ones contained in $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ respectively, and look for the other two.

- Once we identify a subgroup of index 3 in $\mathrm{Gal}(\mathbb{Q}(\zeta_7, \zeta_{13}))$, we wish to find its fixed field in $\mathbb{Q}(\zeta_7, \zeta_{13})$, then use the Dedekind criterion and the Minkowski bound to find the number of generators of the ideal class group, and hope to find a relation between these generators to compute the same.

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$\tau_{\bar{5}}$ of order 4

$$\begin{aligned}\tau_{\bar{5}} : \mathbb{Q}(\zeta_{13}, \zeta_7) &\to \mathbb{Q}(\zeta_{13}, \zeta_7)\\ \zeta_7 &\mapsto \zeta_7\\ \zeta_{13} &\mapsto \zeta_{13}^5\\ (1 &\mapsto 1)\end{aligned}$$

We now define $G' \leqslant G := \mathrm{Gal}(\mathbb{Q}(\zeta_7, \zeta_{13}))$ to be $G' := \langle\, \sigma_{\bar{6}}, \gamma_{(\bar{2},\bar{3})}, \tau_{\bar{5}} \,\rangle$

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We wish to find the fixed field $K' := \mathbb{Q}(\zeta_{91})^{G'}$ which is a degree 3 abelian extension. We let $a := \zeta_7$ and $b := \zeta_{13}$. We make use of the software **SageMath** for some of the computations.

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We observe that the element $\beta := \sum_{\sigma \in G'} \sigma(ab) = a^2b^3 + a^4b^9 + ab$ is fixed by $\langle \gamma_{(\bar{2},\bar{3})} \rangle$.

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[7] We are not using any information from **Sage** which is unknown to us. Finding the minimal polynomial of an algebraic integer θ over $\mathbb{Q}$ can be done using the *regular representation* of $\mathbb{Q}(\theta)$ over $\mathbb{Q}$ with respect to a $\mathbb{Q}-$basis of $\mathbb{Q}(\theta)$. The minimal polynomial of θ would then be the characteristic polynomial of the matrix representing θ.

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Thus, $\mathbb{Q}(\zeta_7, \zeta_{13})^{\langle \gamma_{(\bar{2},\bar{3})} \rangle} = \mathbb{Q}(a^2b^3 + a^4b^9 + ab)$.

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We now find the minimal polynomial of the element
$\beta_2 := a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b$ on **Sage** to be of degree 12.

As $\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle$ is of index 12 in G, we conclude that
$\mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle} = \mathbb{Q}(\beta_2) = \mathbb{Q}(a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b)$.

We finally find the sub field of $\mathbb{Q}(\beta_2)$ which is fixed by the last remaining generator $\tau_{\bar{5}}$ of G'.

The conjugates of β_2 under $\tau_{\bar{5}}$ are:
$a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b$, $a^2b^2 + a^4b^6 + ab^5 + a^5b^2 + a^3b^6 + a^6b^5$,
$a^2b^{10} + a^4b^4 + ab^{12} + a^5b^{10} + a^3b^4 + a^6b^{12}$, and
$a^2b^{11} + a^4b^7 + ab^8 + a^5b^{11} + a^3b^7 + a^6b^8$.

Therefore, let
$\beta_3 := a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b + a^2b^2 + a^4b^6 + ab^5 + a^5b^2 + a^3b^6 + a^6b^5 + a^2b^{10} + a^4b^4 + ab^{12} + a^5b^{10} + a^3b^4 + a^6b^{12} + a^2b^{11} + a^4b^7 + ab^8 + a^5b^{11} + a^3b^7 + a^6b^8$.

We find the minimal polynomial of β_3 over $\mathbb{Q}$ on **Sage** to be $X^3 - X^2 - 30X + 64$.

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We finally have the following tower diagram:

$$
\begin{array}{c}
\mathbb{Q}(\zeta_{91}) \\
\vdots \ \deg 3 \\
\mathbb{Q}(a^2b^3 + a^4b^9 + ab) = \mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})} \rangle} \\
\vdots \ \deg 2 \\
\mathbb{Q}(\beta_2) = \mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle} \\
\vdots \ \deg 4 \\
\mathbb{Q}(\beta_3) = \mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\,\bar{3})},\, \sigma_{\bar{6}},\, \tau_{\bar{5}} \rangle} \\
\vdots \ \deg 3 \\
\mathbb{Q}
\end{array}
$$

Applying some standard techniques and using the fact that $K' \subset \mathbb{Q}(\zeta_{91}) = \mathbb{Q}(\zeta_{13}, \zeta_7)$, we get the following bound on the discriminant

$$d_{\mathbb{Q}(\beta_3)/\mathbb{Q}} \leq 7^2 \cdot 13^2$$

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We now wish to calculate the ideal class group $\mathrm{Cl}(K')$ of the number field K'. The ideal class group $\mathrm{Cl}(K) := \mathscr{I}_K/\mathscr{P}_K$, where $\mathscr{P}_K$ denotes the subgroup of all principal fractional ideals of K, measures by how far the number ring $\mathcal{O}_K$ of the number field K fails from being a principal ideal domain.

We use **Sage** for computing the norms. For example, we can get the elements $\beta_3 \pm i$ in $\mathcal{O}_{K_3}$ for which the prime factorization of their norms $N_{K_3/\mathbb{Q}}(\beta_3 \pm i)$, for $1 \leq i < 100$ have a factor of 17 in them using the following code.

```
K.<b> = NumberField(x^3 -x^2- 30*x + 64)
l=[]
l2=[]
for i in range(1,100):
  I2=K.fractional_ideal(b+i)
  I3=K.fractional_ideal(b-i)
  if 17.divides(int(I2.norm())) == True:
    l.append((b+i, factor(I2.norm())))
    l2.append(factor(I2.norm()))
  if 17.divides(int(I3.norm())) == True:
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    #print(l2)
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We can now use the above information along with the below proposition to get more information of the relation between the generators of the ideal class group.

Proposition

Let K be a number field. Suppose that α is a primitive for K. i.e., $K = \mathbb{Q}(\alpha)$. If there exists a unique $0 \leq r < q$ such that $q \mid N_{K/\mathbb{Q}}(\alpha + r)$, then there exists a unique prime $\mathfrak{q} \in \mathcal{O}_K$ such that $\mathfrak{q}|q$ and $f(\mathfrak{q}|q) = 1$.

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